

DPC Technology Watch Report: Digital Forensics and Preservation



Callum McKean, Digital Lead Curator, Contemporary Archives and Manuscripts, The British Library

Leontien Talboom, Technical Analyst, Cambridge University Library

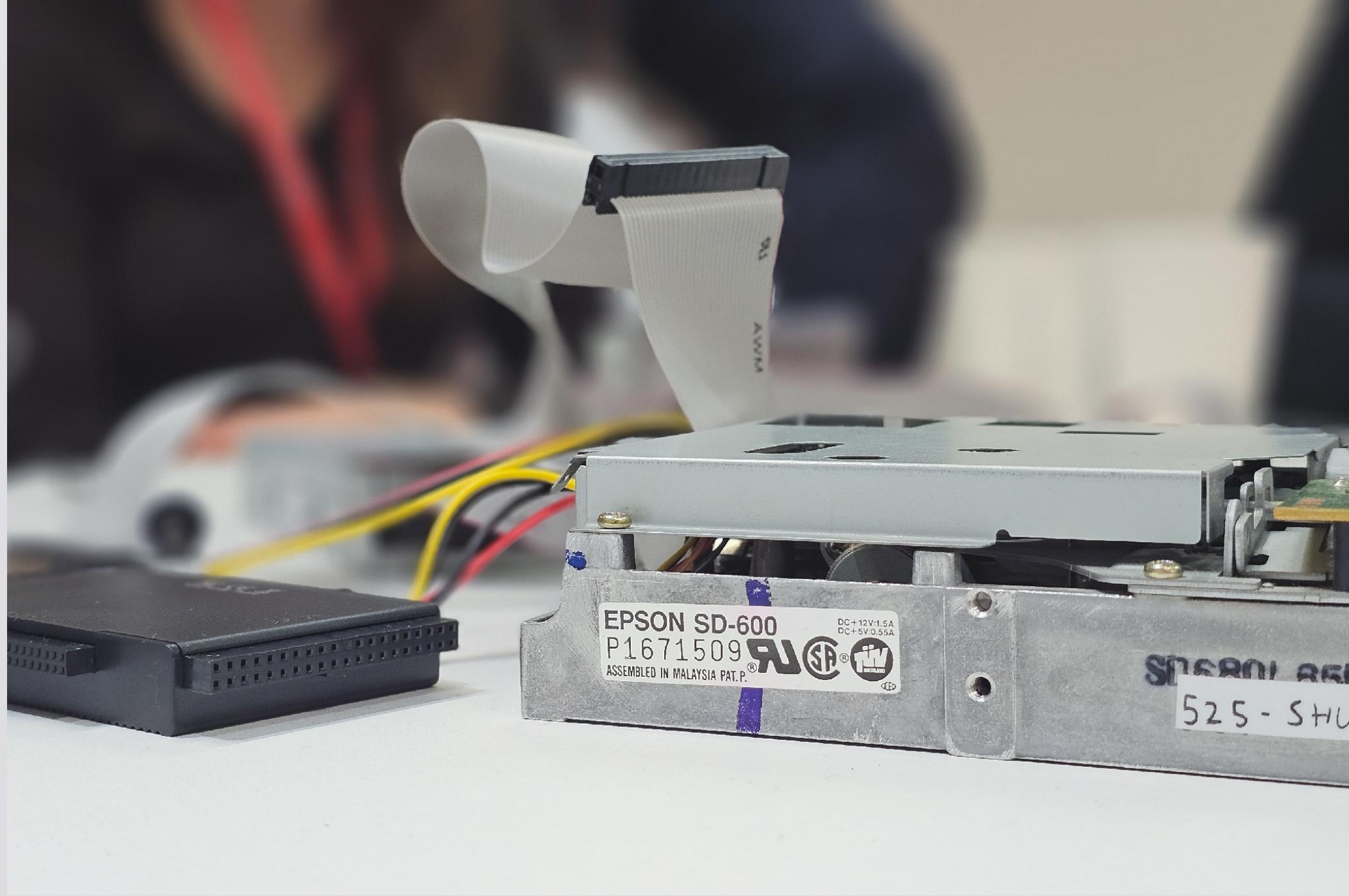
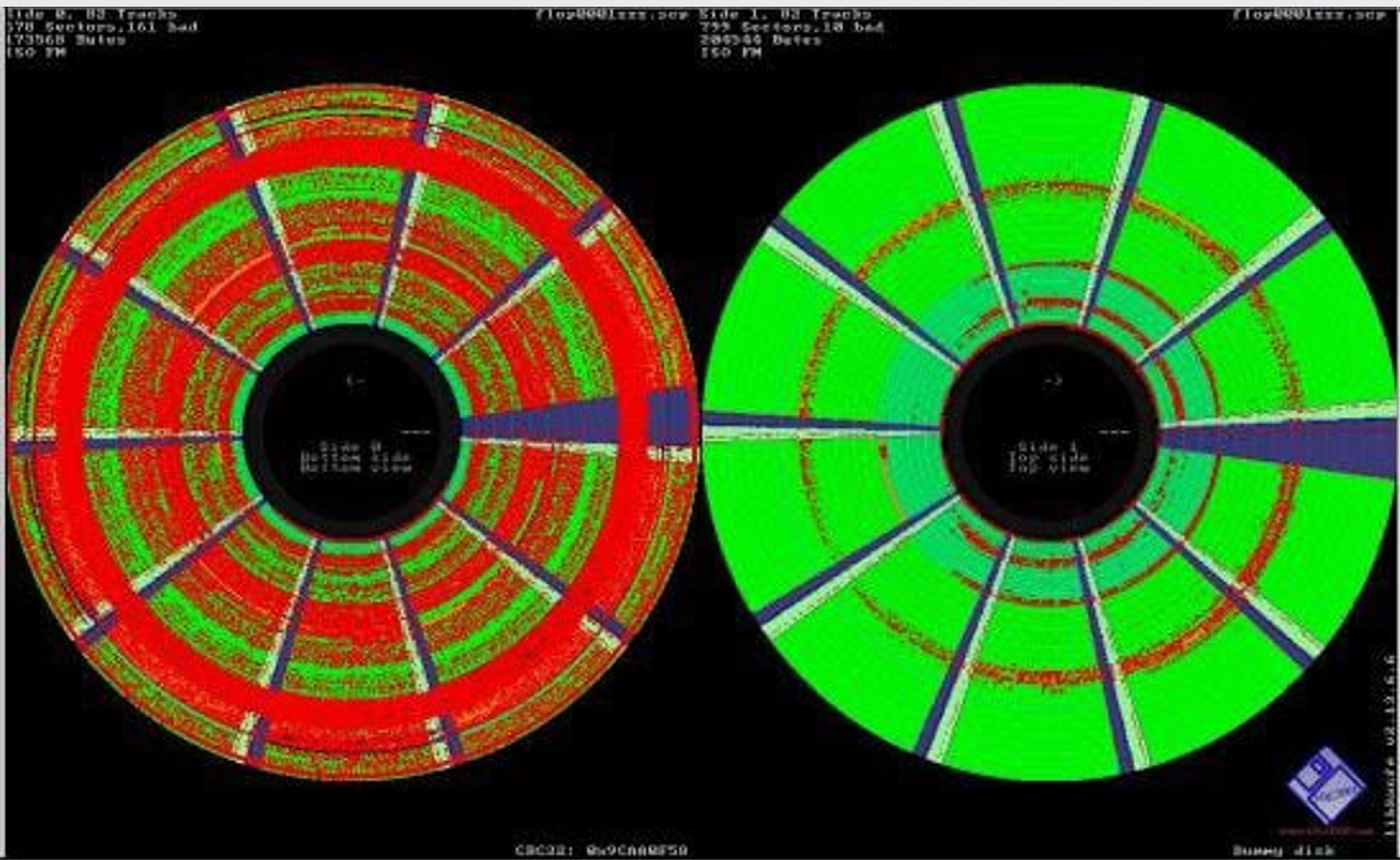
Introduction

This report is an updated version of the 2012 DPC Technology Watch Report on Digital Forensics. The original report was written at a point when digital forensics was only beginning to be adopted within digital preservation practice and therefore focused on defining the field and justifying its relevance to the preservation and representation of personal digital archives.

Since 2012, digital forensics has become embedded in routine practice across our community. Many practitioners now apply forensic methods as part of their day-to-day work. At the same time, the nature of born-digital material has changed significantly, with increased use of personal devices, cloud services, and platform-based systems. This updated report reflects the maturation of the field. It retains useful historical context from the original report, while updating definitions, removing sections that are no longer relevant, and expanding practice-focused guidance. A key change is a stronger emphasis on experience, context, and use, rather than solely on technical, evidentiary concerns or speculative future applications.

Intended Audience

The report is intended for professionals working in digital preservation and cultural heritage who are interested in implementing or refining digital forensics workflows within a preservation setting. Similar to the original report, the focus will be on personal digital archives.



Outline

- Section 1 - Introduction and History of Digital Forensics within Digital Preservation: This section updates the historical narrative to reflect developments since 2012. Earlier definitions of digital forensics are revised to better align with preservation goals, focusing less on law-enforcement models and more on understanding and experiencing digital material in context.
- Section 2 - Digital Forensics in a Broader Context: This section expands the scope beyond traditional storage media to include smartphones, cloud-based services, and platform-dependent content. It also acknowledges related communities, such as retro-computing, software preservation and law enforcement, as contributors to forensic knowledge and practice.
- Section 3 - Legal and Ethical Considerations: This section significantly expands discussion of legal and ethical issues. It focuses on consent, donor agreements, personal data, and professional responsibility, emphasising that cultural heritage institutions have different goals and obligations than law enforcement or the courts.
- Section 4 - Workflows for Digital Forensics: This is the core section of the updated report. It replaces earlier justificatory content with practical guidance on forensic workflows, decision-making, and trade-offs. The focus is on doing the best possible work with available carriers and tools, and on retaining as much context and experiential information as possible.
- Section 5 - Uses of Forensically Captured Material: This section highlights how forensic data is used by digital humanities scholars and other researchers, covering both large-scale computational analysis and close reading approaches. Barriers to wider reuse, including legacy formats and personal data, are also addressed.
- Section 6 - Looking to the Future: This section identifies areas for future focus, including preventing large-scale data loss, engaging with creators earlier, responding to new formats, and supporting research use. It also addresses the loss of computing materiality and the importance of repair and preservation communities. This section will try to avoid speculation about future technical innovation, and will instead base its analysis and recommendations on current issues which threaten our ability to collect, preserve and make this material available, as well as possible practical approaches to mitigating these risks.



Read and comment on the draft Report